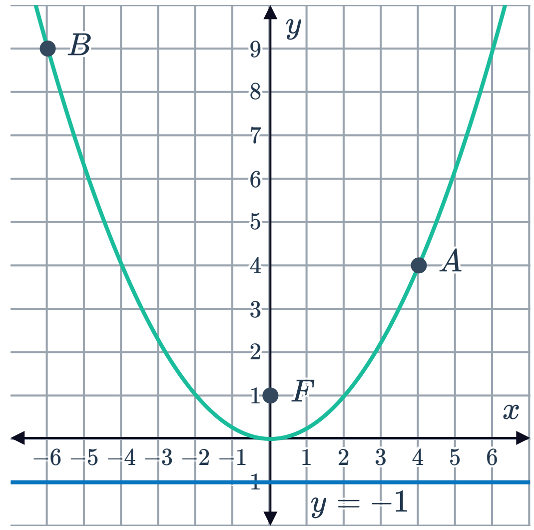
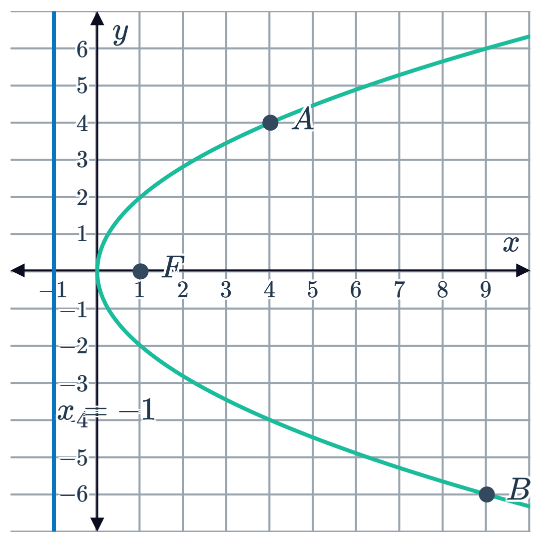
Key features of parabolas

1 The graph of $x^2 = 4y$ has been plotted, along with points $A(4,4)$ and $B(-6,9)$. The point $F(0,1)$ is called the focus and the line $y=-1$ is called the directrix.



- a Find the distance between points A and F.
- b Find the perpendicular distance between point A and the directrix.
- c Find the distance between points B and F.
- d Find the perpendicular distance between point B and the directrix.

2 The graph of $y^2 = 4x$ has been plotted, along with points $A(4,4)$ and $B(9,-6)$. The point $F(1,0)$ is called the focus and the line $x=-1$ is called the directrix.



- a Find the distance between points A and F.
- b Find the perpendicular distance between point A and the directrix.
- c Find the distance between points B and F.
- d Find the perpendicular distance between point B and the directrix.

3 For each of the following parabolas:

- i Does the parabola open upwards or downwards?
- ii State the coordinates of its vertex.
- iii Solve for a, the focal length of the parabola.
- iv State the coordinates of the focus.
- v State the equation of its directrix.

a $(x-2)^2 = 12(y-4)$

b $(x+4)^2 = 6(y+2)$

c $(x-3)^2 = -16(y+2)$

d $y = x^2 + 3$

4 For each of the following parabolas:

i Does the parabola open to the left or to the right?

ii State the coordinates of its vertex.

iii Solve for a, the focal length of the parabola.

iv State the coordinates of the focus.

v State the equation of its directrix.

a $(y-3)^2 = 12x$

b $y^2 = 16(x-2)$

c $(y-3)^2 = -16(x-2)$

d $(y+4)^2 = -12(x-2)$

5 For each of the following parabolas:

i State the coordinates of its vertex.

ii State the equation of its axis of symmetry.

iii Solve for a, the focal length of the parabola.

iv State the coordinates of the focus.

v State the equation of its directrix.

a $x^2 = 16y$

b $x^2 = -\frac{1}{3}y$

c $y^2 = -20x$

d $y^2 = -\frac{1}{5}x$

6 The parabola is represented by the equation $(x-2)^2 = 8(y+1)$.

a Find the coordinates of the vertex.

b Find the coordinates of the focus.

c Find the equation of the directrix.

d Find the equation of the axis of symmetry.

e State the domain in interval notation.

f State the range in interval notation.

7 The parabola is represented by the equation $(y-3)^2 = 8(x+1)$.

a Find the coordinates of the vertex.

b Find the coordinates of the focus.

c Find the equation of the directrix.

d Find the equation of the axis of symmetry.

e State the domain in interval notation.

f State the range in interval notation.

8 Two points on the parabola $x^2 = -12(y+5)$ are each 5 units away from the focus of the parabola.

a Find the x-coordinates of the points.

b State the coordinates of the two points.

9 Two points on the parabola $y^2 = -20(x+4)$ are each 8 units away from the focus of the parabola.

a Find the y-coordinates of the points.

b State the coordinates of the two points.

10 The point $\left(k^2 - 28k - 35, 2k - 1\right)$ lies on the curve $x = y^2$. Find the values of k.

Graphs of parabolas

11 Consider the parabola $x^2 = -16y$.

a Does the parabola open upwards or downwards?

b State the coordinates of its vertex.

c Solve for a, the focal length of the parabola.

d State the coordinates of the focus.

e State the equation of its directrix.

f Sketch the graph of the parabola, along with its directrix and focus.

12 Consider the following graph of $x^2 = 4y$:

a State the coordinates of its focus.

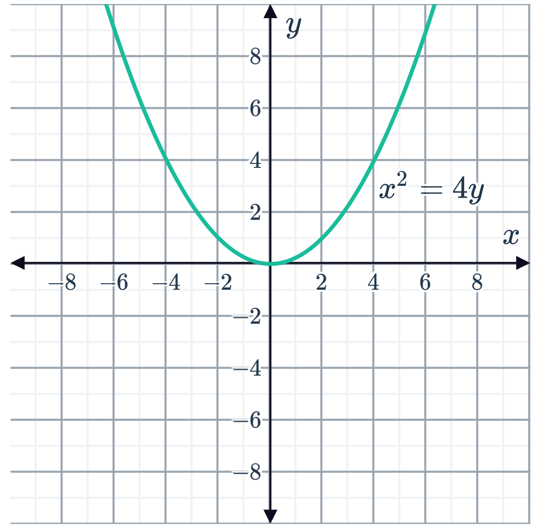
b State the equation of its directrix.

c The graph of $x^2 = 4y$ is shifted 2 units to the right. Sketch the resulting graph.

d State the equation of the graph from part (c).

e State the coordinates of the focus of the transformed graph.

f State the equation of the directrix of the transformed graph.



13 Consider the parabola with equation $y^2 = -8x$.

a State the equation of its directrix.

b Sketch the graph of the parabola and its directrix.

c Two points on the parabola are each 4 units away from the focus of the parabola. Solve for the y-coordinates of the points.

d Now, state the coordinates of the two points.

14 Consider the parabola $x = -\left(y-4\right)^2 + 1$.

a Does the parabola open to the left or to the right?

b Find the coordinates of the vertex.

c Sketch the graph of the parabola.

15 Consider the parabola $\left(x+2\right)^2 = -12\left(y-4\right)$.

a Does the parabola open upwards or downwards?

b State the coordinates of its vertex.

c Solve for a, the focal length of the parabola.

d State the coordinates of the focus.

e State the equation of its directrix.

f Sketch the graph of the parabola, along with its directrix and focus.

16 Consider the parabola $\left(x+4\right)^2 = 6\left(y+3\right)$.

a Does the parabola open upwards or downwards?

b State the coordinates of its vertex.

c Solve for a, the focal length of the parabola.

d State the coordinates of the focus.

e State the equation of its directrix.

f Sketch the graph of the parabola, along with its directrix and focus.

g The point $(-10, 3)$ lies on the parabola. Find the distance of this point from the focus.

Equations of parabolas

17 State the equation of each of the following parabolas based on the given description:

a A parabola has its vertex at the origin. Its focus lies on the y-axis, 4 units above the vertex.

b A parabola has a focus at $(0, 3)$ and a vertex at the origin.

c A parabola has a focus at $(-6, 0)$ and a vertex at the origin.

d A parabola has a focus at $\left(-\frac{1}{2}, 0\right)$ and a vertex at the origin.

e A parabola with a vertex at the origin opens upward and passes through the point $(4, 8)$.

f A parabola with a vertex at the origin is symmetric about the y-axis and passes through the point $(7, 5)$.

g A parabola has a focus at $(6, 4)$ and a directrix of $y = -4$.

h A parabola has a focus at $(5, 8)$ and a directrix of $x = -5$.

18 A parabola has its vertex at the origin. It has a focal length measuring $a = 4$ units, and its focus lies on the x-axis.
State both possible equations of the parabola.

19 A parabola has its vertex at the origin, and it passes through the point $P(-4, -16)$.

a By finding the focal length a or otherwise, state the equation of the parabola if its axis of symmetry is $x = 0$.

b Sketch the graph of the parabola.

20 A parabola has its vertex at the point $(3, 2)$, and it passes through the point $P(1, -2)$.

a Sketch the graph of the parabola.

b State the equation of the parabola.

Applications

21 When an object is thrown into the air, its height above the ground is given by the equation

$$h = 193 + 24s - s^2$$

where s is its horizontal distance from where it was thrown.

a Find the value of s, at the point when it reaches its greatest height above the ground.

b Find the maximum height reached by the object.

22 The height at time t of a ball thrown upwards is given by the equation $h = 59 + 42t - 7t^2$.

a How long does it take the ball to reach its maximum height?

b Find the maximum height reached by the ball.